

STANKIN University
MSc in Artificial Intelligence and Big Data

Reliable Vision

Confidence Calibration and View Agreement
for Image Classification under Degradation

Research Project Report

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Code and experiment notebook

<https://colab.research.google.com/drive/1Az1TWZLeuRQzrPp5SM40-Z39nk4BpHo3?usp=sharing>

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Abstract

This project examines whether an image classifier can reduce mistakes by referring uncertain predictions for human review. A small convolutional neural network was trained on CIFAR-10, with separate data for checkpoint selection, confidence calibration, and review-rule development. Two rules were compared: calibrated maximum probability, and that probability multiplied by prediction agreement across three transformed views.

The classifier achieved 84.34% accuracy on the clean test set. Temperature scaling reduced clean test expected calibration error from 0.0241 to 0.0109. The confidence rule answered 73.65% of clean test images at 94.57% accuracy; the agreement rule answered 71.98% at 94.98% accuracy. Agreement improved error ranking and accepted accuracy under Gaussian noise, while reducing coverage. It did not provide consistent improvement under Gaussian blur. The results show that stable predictions can still be wrong and that a review threshold developed on clean images does not ensure the same accuracy after image degradation. These findings come from one training seed and a controlled synthetic-corruption study.

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1 Project aim

The aim was to build and evaluate an image classification pipeline that can either return a prediction or request human review. The project focuses on the reliability of predictions, rather than accuracy alone.

The main question was:

Can confidence calibration and prediction agreement reduce errors among accepted predictions, and how does this depend on image quality?

A second question was whether a rule selected on clean images remains useful when the same type of images becomes blurry or noisy.

Selective classification describes this trade-off between answering more inputs and reducing errors among the answers [3]. Temperature scaling provides an established way to adjust confidence scores [2]. This project applies these ideas in a reproducible experiment and compares a confidence baseline with a simple agreement-based score.

1.1 Project contribution

The project combines:

- a trained classifier with distinct training, validation, calibration, and policy-development splits;
- a reproducible implementation of two review rules;
- a comparison on clean images and four controlled degradations;
- saved predictions, fixed thresholds, and uncertainty intervals;
- an analysis of the different behaviour under noise and blur.

The contribution is the implementation and empirical comparison. The score combines existing signals; a new general theory or a state-of-the-art classification method is not claimed.

2 Dataset and experimental protocol

2.1 CIFAR-10

CIFAR-10 contains 60,000 RGB images of size 32×32 , divided into ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. Its original split contains 50,000 training images and 10,000 test images [1].

The original training portion was divided into four disjoint, class-balanced subsets with stratified sampling and seed 42. The official test set was kept separate.

2.2 Order of experiments

The experiment followed this order:

1. Train the classifier and select the checkpoint with the highest validation accuracy.
2. Fit temperature scaling using only calibration images.

Table 1: Data splits and their roles.

Subset	Images	Per class	Use
Training	35,000	3,500	Learn classifier weights
Validation	5,000	500	Select the best training checkpoint
Calibration	5,000	500	Fit the temperature parameter
Policy development	5,000	500	Select review thresholds and examine failures
Final test	10,000	1,000	Evaluate frozen methods

3. Select the confidence threshold using clean policy images.
4. Examine the confidence rule on degraded policy images.
5. Add an agreement score, select its threshold on clean policy images, and examine its behaviour under degradation.
6. Freeze the checkpoint and both rules before final test evaluation.

The agreement method was added during development after observing failures on the policy set. This was a development decision; test labels were not used to choose the method or its threshold. Each review threshold was selected on clean policy images only.

The final evaluation uses five versions of each of the 10,000 test images. These are 50,000 condition-specific observations of 10,000 underlying images, rather than 50,000 independent images. Results are therefore reported separately by condition.

3 Classifier and training

3.1 Network architecture

The baseline, **SmallCNN**, contains three convolutional blocks. Each block uses two 3×3 convolutions with padding one. Each convolution is followed by batch normalization and ReLU. A 2×2 max-pooling layer follows each block. Convolutional bias terms are disabled.

Table 2: Baseline architecture. Shapes exclude the batch dimension.

Stage	Operations	Output shape
Input	Normalized RGB image	$3 \times 32 \times 32$
Block 1	Two convolutions, 32 channels; pooling	$32 \times 16 \times 16$
Block 2	Two convolutions, 64 channels; pooling	$64 \times 8 \times 8$
Block 3	Two convolutions, 128 channels; pooling	$128 \times 4 \times 4$
Classifier	Flatten; linear 2,048 to 256; ReLU	256
Output	Dropout 0.3; linear 256 to 10	10 logits

The architecture has 814,570 trainable parameters. It returns raw class scores, or logits; softmax is applied separately.

3.2 Training settings

Pixel values were converted to $[0, 1]$ and normalized using channel mean 0.5 and standard deviation 0.5:

$$x_{\text{norm}} = \frac{x - 0.5}{0.5}. \quad (1)$$

Training augmentation consisted of a random 32×32 crop with padding four and a random horizontal flip. Validation, calibration, and policy images did not receive training augmentation.

Table 3: Training and execution settings.

Setting	Value
Training seed	42
Epochs	15
Batch size	128
Loss	Multiclass cross-entropy
Optimizer	AdamW
Initial learning rate	10^{-3}
Weight decay	10^{-4}
Scheduler	Cosine annealing, $T_{\text{max}} = 15$
Data-loader workers	2
Checkpoint criterion	Highest validation accuracy
Selected epoch	14
Best validation accuracy	85.36%
Execution environment	Google Colab with CUDA
Reported PyTorch version	2.11.0+cu130

One model was trained. Exact repeatability across different hardware and library versions was not established.

4 Confidence calibration and review rules

4.1 Temperature scaling

Let $z_k(x)$ denote the logit for class k . Temperature scaling uses a positive scalar T :

$$p_T(k \mid x) = \frac{\exp(z_k(x)/T)}{\sum_{j=1}^{10} \exp(z_j(x)/T)}. \quad (2)$$

The fitted temperature minimizes calibration-set negative log-likelihood:

$$T^* = \arg \min_T \left[-\frac{1}{n_{\text{cal}}} \sum_{i=1}^{n_{\text{cal}}} \log p_T(y_i \mid x_i) \right]. \quad (3)$$

The implementation optimized $\ell = \log T$ using L-BFGS, with learning rate 0.1, a maximum of 100 iterations, and a strong-Wolfe line search. In the objective, ℓ was clipped to $[-4, 4]$ before exponentiation. If the fitted calibration loss worsened or became non-finite, the implementation reverted to $T = 1$.

The fitted value was $T = 1.1492$ to four decimal places. Calibration-set NLL fell from 0.4411 to 0.4359. These are fitting results, not independent test evidence.

Because $T > 0$ divides all logits by the same value, it does not change the predicted class:

$$\hat{y}(x) = \arg \max_k z_k(x) = \arg \max_k p_T(k \mid x). \quad (4)$$

Confidence calibration changes probabilities, not classification accuracy or model weights [2].

4.2 Confidence rule

The first score is the calibrated maximum probability:

$$c(x) = \max_k p_T(k \mid x). \quad (5)$$

An image is accepted when

$$g_{\text{conf}}(x) = \mathbf{1}[c(x) \geq \tau_{\text{conf}}]. \quad (6)$$

Otherwise it is referred for human review.

4.3 Confidence multiplied by view agreement

The second rule examines three versions of each input:

1. a horizontal flip;
2. Gaussian blur with a 9×9 kernel and $\sigma = 0.5$;
3. Gaussian noise with standard deviation 0.02 in $[0, 1]$ pixel space, followed by clipping.

For a degraded condition, these transformations are applied to the already degraded input. They are not applied to an unavailable clean reference image.

Agreement is the fraction of transformed views whose predictions match the original prediction:

$$a(x) = \frac{1}{3} \sum_{v=1}^3 \mathbf{1}[\hat{y}(t_v(x)) = \hat{y}(x)]. \quad (7)$$

The score and acceptance rule are

$$s(x) = c(x)a(x), \quad (8)$$

$$g_{\text{agr}}(x) = \mathbf{1}[s(x) \geq \tau_{\text{agr}}]. \quad (9)$$

Agreement takes values in $\{0, 1/3, 2/3, 1\}$. The product $s(x)$ is a heuristic ranking score; it is not itself a calibrated probability of correctness. Accepted predictions retain the original class label. The transformed predictions are not averaged to change that label.

The confidence rule needs one forward pass. The agreement rule needs four: one original image and three views. Wall-clock latency was not benchmarked.

4.4 Threshold selection

For each method, thresholds were searched on the grid $\{0, 0.01, \dots, 1\}$. The selected threshold maximized clean policy coverage subject to accepted accuracy of at least 95% and at least 500 accepted images:

$$\tau^* \in \arg \max_{\tau} \hat{C}_{\text{policy}}(\tau) \quad \text{subject to} \quad \hat{A}_{\text{policy}}(\tau) \geq 0.95, \quad n_{\text{accepted}}(\tau) \geq 500. \quad (10)$$

Table 4: Thresholds selected on clean policy images.

Rule	Threshold	Accepted	Coverage	Accepted accuracy
Confidence	0.75	3,647	72.94%	95.04%
Confidence + agreement	0.70	3,591	71.82%	95.15%

Both rules met the empirical development criterion. This criterion does not guarantee 95% accuracy on new images or under degradation. If no threshold met both constraints, the code would refer all images for review.

5 Image degradation

Four synthetic conditions were evaluated in addition to clean images:

Table 5: Controlled image degradations.

Condition	Operation
Blur 1	Gaussian blur; 9×9 kernel; $\sigma = 1$
Blur 2	Gaussian blur; 9×9 kernel; $\sigma = 2$
Noise 0.05	Add Gaussian noise with standard deviation 0.05; clip to $[0, 1]$
Noise 0.10	Add Gaussian noise with standard deviation 0.10; clip to $[0, 1]$

Noise was generated in pixel space and the result was normalized again using the classifier’s preprocessing. For pixel x , the corruption is

$$\tilde{x} = \text{clip}(x + \epsilon, 0, 1), \quad \epsilon \sim \mathcal{N}(0, \sigma_n^2). \quad (11)$$

Blur and noise were evaluated separately; they were not combined. Ground-truth labels were kept unchanged. Severe degradation can remove information needed to recognize an image, so the experiment does not assume that every corrupted example remains easy for a person to classify.

Policy corruption and view-noise seeds were 142 and 242. Final test seeds were 1,042 and 2,042. Generators were reset for each condition. With a fixed loader order, the two noise levels use corresponding Gaussian draws scaled to different standard deviations. These are custom synthetic corruptions, not an evaluation of the official CIFAR-10-C dataset.

6 Evaluation metrics

For n images, correctness is $r_i = \mathbf{1}[\hat{y}_i = y_i]$, and g_i indicates acceptance.

6.1 Accuracy, coverage, and accepted errors

Overall accuracy, coverage, and accepted accuracy are

$$A = \frac{1}{n} \sum_i r_i, \quad (12)$$

$$C = \frac{1}{n} \sum_i g_i, \quad (13)$$

$$A_{\text{accepted}} = \frac{\sum_i g_i r_i}{\sum_i g_i}. \quad (14)$$

Accepted accuracy is undefined if no image is accepted. The accepted error rate is $1 - A_{\text{accepted}}$, and the accepted error count is $\sum_i g_i(1 - r_i)$.

The full-answer baseline accepts every image. Rejected images are not counted as correct classifications. They require a later decision by a person; actual human review was not evaluated.

6.2 Calibration metrics

Negative log-likelihood is

$$\text{NLL} = -\frac{1}{n} \sum_i \log p_T(y_i | x_i). \quad (15)$$

Expected calibration error uses 15 equal-width confidence bins:

$$\text{ECE} = \sum_{b=1}^{15} \frac{|B_b|}{n} |\text{accuracy}(B_b) - \text{confidence}(B_b)|. \quad (16)$$

Empty bins contribute zero. Confidence one is included in the final bin. ECE depends on binning and is not a complete measure of calibration.

The multiclass Brier score uses the sum over all ten classes:

$$\text{Brier} = \frac{1}{n} \sum_i \sum_{k=1}^{10} (p_T(k | x_i) - \mathbf{1}[y_i = k])^2. \quad (17)$$

NLL, ECE, and Brier score are better when lower.

6.3 Error detection and uncertainty intervals

Error-detection AUROC treats incorrect predictions as the positive class. The error score is $1 - c(x)$ or $1 - s(x)$. Higher AUROC indicates better ranking of incorrect predictions, without fixing an acceptance threshold.

Accepted accuracy is accompanied by an approximate 95% Wilson binomial interval [4]. For m accepted images, observed accuracy $\hat{p}$, and $z = 1.95996$, its center and half-width are

$$\mu_W = \frac{\hat{p} + z^2/(2m)}{1 + z^2/m}, \quad (18)$$

$$h_W = \frac{z}{1 + z^2/m} \sqrt{\frac{\hat{p}(1 - \hat{p})}{m} + \frac{z^2}{4m^2}}. \quad (19)$$

The interval is $[\mu_W - h_W, \mu_W + h_W]$, clipped to $[0, 1]$.

These intervals describe binomial sampling uncertainty within each condition. They do not measure variation across training seeds, provide simultaneous coverage over all conditions, or establish statistical significance between methods.

7 Final test results

The final test used the same trained classifier, fitted temperature, and frozen thresholds in every condition. The saved checkpoint’s SHA-256 hash and rule configuration were recorded before inference. Prediction caches allow later analysis without changing the model or rerunning the evaluation.

In Table 6, ALL answers every image, CONF uses confidence alone, and CONF+AGR uses confidence multiplied by agreement. Coverage and accepted accuracy are percentages. Each condition contains 10,000 images.

Table 6: Final test results for the frozen rules.

Condition	Rule	Coverage	Accuracy	95% CI	Accepted	Errors	AUROC
Clean	ALL	100.00	84.34	83.61–85.04	10,000	1,566	–
	CONF	73.65	94.57	94.03–95.06	7,365	400	0.8784
	CONF+AGR	71.98	94.98	94.46–95.47	7,198	361	0.8786
Blur 1	ALL	100.00	37.84	36.89–38.80	10,000	6,216	–
	CONF	31.72	57.76	56.03–59.46	3,172	1,340	0.6742
	CONF+AGR	36.03	54.48	52.85–56.10	3,603	1,640	0.6575
Blur 2	ALL	100.00	24.00	23.17–24.85	10,000	7,600	–
	CONF	16.91	34.54	32.31–36.83	1,691	1,107	0.5867
	CONF+AGR	21.44	33.02	31.06–35.04	2,144	1,436	0.5968
Noise 0.05	ALL	100.00	63.40	62.45–64.34	10,000	3,660	–
	CONF	55.44	81.33	80.28–82.34	5,544	1,035	0.7898
	CONF+AGR	51.60	87.42	86.49–88.30	5,160	649	0.8342
Noise 0.10	ALL	100.00	34.08	33.16–35.01	10,000	6,592	–
	CONF	44.10	45.53	44.07–47.01	4,410	2,402	0.6678
	CONF+AGR	35.24	57.52	55.88–59.14	3,524	1,497	0.7357

7.1 Clean images

The classifier correctly classified 8,434 of 10,000 clean images. The confidence rule accepted 7,365 images and referred 2,635. Among accepted images, 400 predictions were wrong.

The agreement rule accepted 7,198 images and referred 2,802. It made 361 accepted errors, giving 94.98% accepted accuracy. Compared with confidence alone, it answered 167 fewer images and made 39 fewer accepted errors. This is a different accepted subset, not evidence that the underlying classifier was improved.

The clean test point estimates, 94.57% and 94.98%, were both below the 95% development target. Both reported intervals include 95%. The experiment therefore does not demonstrate a guaranteed 95% clean test accuracy for accepted predictions.

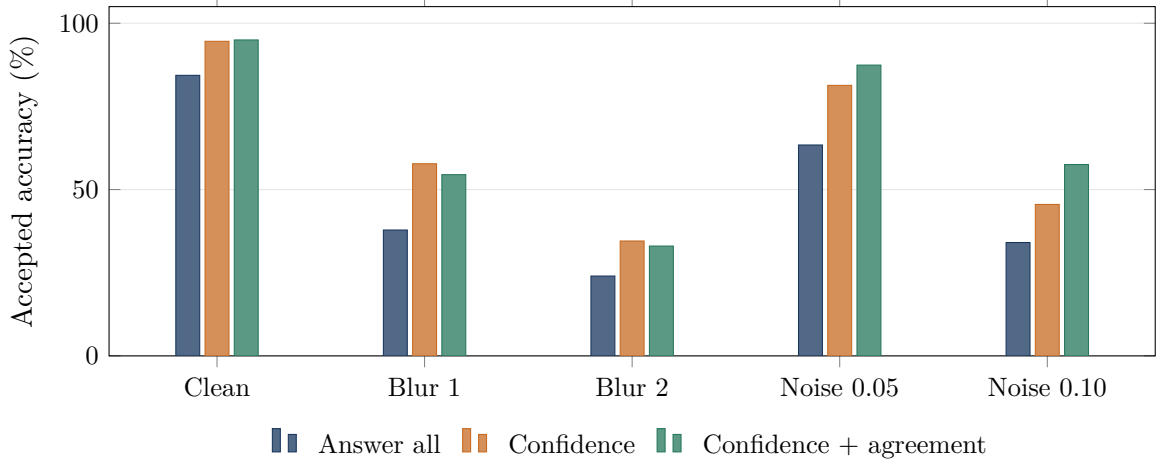


Figure 1: Accuracy among accepted test predictions. Each rule accepts a different subset; coverage is shown separately in Figure 2.

7.2 Gaussian noise

At noise standard deviation 0.05, agreement increased accepted accuracy from 81.33% to 87.42%, a difference of 6.09 percentage points. Coverage fell from 55.44% to 51.60%. Accepted errors fell from 1,035 to 649.

At noise standard deviation 0.10, accepted accuracy increased from 45.53% to 57.52%, while coverage fell from 44.10% to 35.24%. Accepted errors fell from 2,402 to 1,497.

These results combine a change in error ranking with a change in coverage. They do not establish superiority at equal coverage. However, AUROC also increased under both noise conditions: from 0.7898 to 0.8342 and from 0.6678 to 0.7357. This supports better error ranking under noise in this run, without establishing statistical significance.

7.3 Gaussian blur

Under blur $\sigma = 1$, the agreement rule accepted more images but had lower accepted accuracy: 54.48%, compared with 57.76% for confidence alone. Accepted errors increased from 1,340 to 1,640. Its error-detection AUROC also decreased.

Under blur $\sigma = 2$, accepted accuracy was 33.02% with agreement and 34.54% with confidence alone. The agreement rule again answered more images. Although its AUROC was slightly higher, both ranking scores performed poorly.

The selected agreement threshold is lower than the confidence threshold. An image with full agreement can therefore pass the agreement rule at confidence 0.70 even when it fails the confidence rule at 0.75. This explains why agreement can have higher coverage under blur despite multiplying confidence by a value no greater than one.

The results are consistent with an important failure mode: a classifier can make the same wrong prediction across several views. Agreement does not prove that the image was understood.

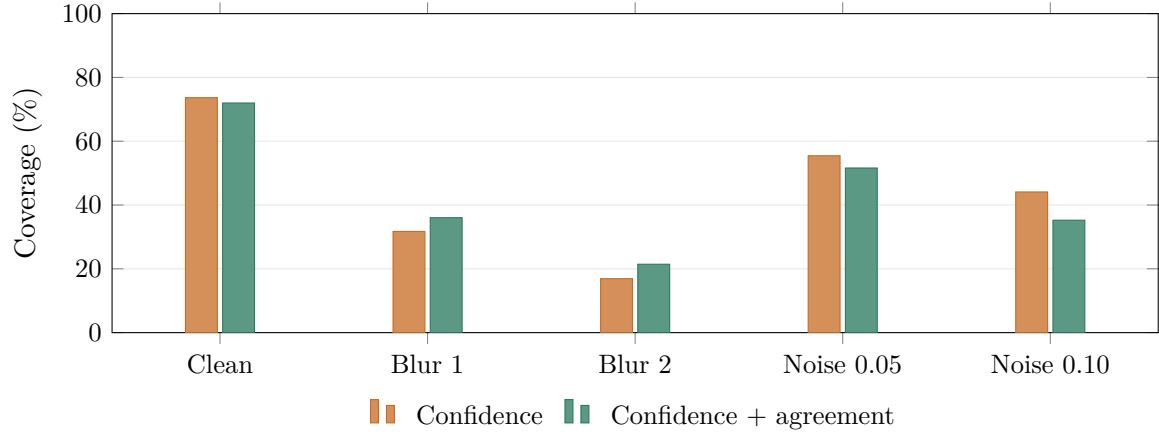


Figure 2: Fraction of test images answered by each frozen rule. The full-answer baseline has 100% coverage in every condition.

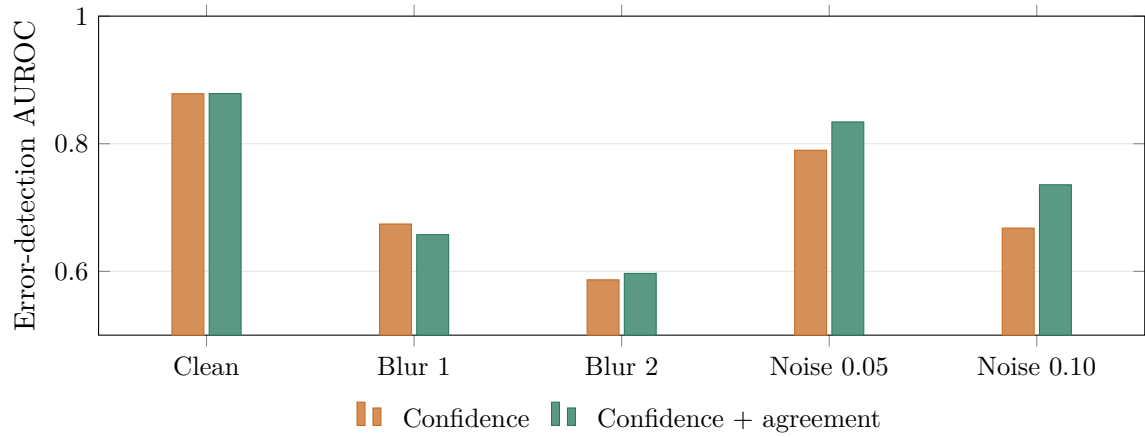


Figure 3: Error ranking on final test images. Larger values are better; 0.5 is the chance-ranking reference.

7.4 Calibration on the final test set

Temperature scaling left classification accuracy unchanged. Table 7 reports calibration metrics before and after applying the temperature fitted on clean calibration data.

Table 7: Final test calibration metrics. All metrics are reported as raw values, not percentages.

Condition	Calibration	NLL	ECE	Brier
Clean	Before	0.4599	0.0241	0.2222
	After	0.4532	0.0109	0.2213
Blur 1	Before	2.4400	0.3043	0.8775
	After	2.2458	0.2620	0.8472
Blur 2	Before	2.8298	0.3611	1.0414
	After	2.6283	0.3089	0.9959
Noise 0.05	Before	1.2205	0.1487	0.5223
	After	1.1464	0.1181	0.5088
Noise 0.10	Before	2.5963	0.3864	0.9691
	After	2.3676	0.3480	0.9313

NLL, ECE, and Brier score decreased in every tested condition. The strongest clean calibration finding is the ECE reduction from 0.0241 to 0.0109. Nevertheless, calibrated ECE remained high under severe noise and blur. A useful clean-data correction did not make the classifier reliable under these shifts.

8 Discussion

8.1 What worked

On clean images, both review rules selected subsets with much higher accuracy than the full-answer baseline, at the cost of sending about one quarter of images for review. Temperature scaling also improved clean confidence calibration.

Under Gaussian noise, prediction agreement provided a useful additional signal. The observed improvement in error AUROC indicates that its benefit was not explained only by the chosen acceptance threshold. The frozen agreement rule still missed many errors under severe noise, so its usefulness is relative to the confidence baseline rather than evidence of reliable deployment.

8.2 What did not work

Blur removed information while leaving some predictions both confident and stable. The agreement score could not consistently separate these errors from correct predictions.

This distinction matters for practical systems. Confidence asks how strongly the classifier prefers its answer. Agreement asks whether that answer changes under the chosen transformations. Neither directly verifies that the answer is correct.

8.3 Use as a prototype

The pipeline could serve as a prototype for routing uncertain images to a review queue in a controlled ten-class task. The implemented experiment evaluates that routing decision, not a

deployed application, a user interface, or the accuracy of a human reviewer.

Before practical use, the system would need task-specific images, evaluation under expected camera conditions, and a measured review workflow. It should not be treated as a general object recognition system or as a solution for detecting all unknown classes.

9 Limitations and further work

The main limitations are:

- **One training seed.** The intervals do not capture variation from retraining the network.
- **One small dataset and architecture.** Results do not establish behaviour on high-resolution images or other models.
- **Synthetic degradation.** Gaussian blur and noise do not cover all real camera failures, compression, motion blur, lighting changes, or domain shifts.
- **Clean-data thresholds.** The 95% policy target was neither a statistical guarantee nor a corruption-specific target.
- **Different coverage.** Fixed-threshold accuracy comparisons use different accepted subsets. Matched-coverage curves and paired statistical comparisons were not computed.
- **Limited transformation choice.** Three views give a coarse agreement score. Even mild transformations can affect valid predictions, while systematic errors can remain stable.
- **No human study or latency benchmark.** Review quality, review cost, and actual runtime were not measured.

Useful extensions would include repeated training seeds, matched-coverage analysis, per-class failure analysis, quality-aware rejection, and evaluation on another dataset. Any new method should be developed on separate development data and assessed on new held-out data, rather than tuned to improve the already observed test results.

10 Conclusion

This project demonstrates a complete experimental pipeline for classification with a human-review option. On clean test images, calibrated confidence achieved 94.57% accepted accuracy at 73.65% coverage. Adding view agreement achieved 94.98% at 71.98% coverage.

Agreement improved error ranking under Gaussian noise but did not consistently help under blur. Temperature scaling improved the reported calibration metrics without correcting the classifier’s predictions. The central finding is that confidence and stability provide useful but incomplete evidence: a stable answer can still be wrong, especially when image degradation removes the information needed for recognition.

A Code and reproducibility

The Google Colab notebook is available at:

<https://colab.research.google.com/drive/1Az1TWZLeuRQzrPp5SM40-Z39nk4BpHo3?usp=sharing>

The project output directory used in the experiments was:

```
/content/drive/MyDrive/Reliable-Vision-Jaber
```

The workflow records split indices, preprocessing and training settings, classifier weights, temperature, review thresholds, policy diagnostics, and final test predictions.

The main saved records include:

- `split_indices.npz` and `config.json`;
- `baseline_seed42_best.pt`;
- `baseline_seed42_calibrated.pt`;
- `baseline_seed42_config.json`;
- `baseline_seed42_history.csv`;
- `baseline_seed42_review_policy.json`;
- `baseline_seed42_consistency_policy.json`;
- `baseline_seed42_frozen_test_config.json`;
- `baseline_seed42_final_test_results.csv`;
- `baseline_seed42_final_test_calibration.csv`.

Condition-specific prediction caches use the pattern

```
baseline_seed42_test_<condition>_predictions.npz.
```

They contain logits, labels, predictions, confidence, view agreement, the combined score, and correctness. A frozen configuration is checked before cached test predictions are reused.

To reproduce the workflow, run the notebook cells in order. The initial cells mount Google Drive and download CIFAR-10. Later cells reuse the dataset, train the classifier, fit calibration, develop review rules, and evaluate the final test set. The exact recorded library versions should be retained when attempting a close reproduction.

The numerical results come from the recorded experiment outputs. Values in tables are rounded; calculations using rounded values may differ slightly from the saved full-precision records. The figures are drawn directly from the final test tables.

References

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